

Mayur Kumar

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PROFILE SUMMARY

Embedded Software Engineer with **2+ years of hands-on experience** developing, debugging, and upstreaming Linux kernel drivers on ARM-based SoC platforms. Strong expertise in kernel subsystems, low-level hardware interaction, and performance-critical driver development for proprietary and platform-specific features.

Proficient in **Linux Kernel 6.6**, with direct experience in **upstream kernel contribution**, adhering to kernel coding standards, review workflows, and long-term maintainability expectations. Solid background in BSP development, boot flow, and system bring-up, including integration across Linux and OpenWrt environments.

Deep understanding of **ARM architecture**, **Trusted Firmware-A (TF-A)**, **TrustZone**, and **Secure Monitor Calls (SMC)**, enabling secure coordination between kernel space and firmware.

SKILLS

ARM Ecosystem	ARM AArch64, ARM TrustZone, GPT, Secure Monitor Call (SMC), Embedded Linux, U-Boot, Device Tree (DT)
Kernel/OS	Linux Kernel (up to 6.6), Kernel Driver Development, OpenWrt, Kernel API, I/O Subsystems
Hardware/SoC	SoC Architecture, I2C/SPI/UART, Memory-Mapped I/O (MMIO), Hardware/Software Interface

WORK EXPERIENCE

Engineer (Full Time) - Airoha July 2024 – Present

- Build the **GUID Partition Table** support in the ARM Trusted Firmware for EMMC flash support.
- Developed, debugged, and contributed **Linux kernel driver patches** to support multiple devices and platform-specific hardware variants.
- Spearheaded integration and compatibility testing of **proprietary kernel modules** for the SoC across **Linux Kernel 6.6/6.12** and **OpenWrt 23.05/24.10/25.12** releases.
- Designed and implemented the **SMC (Secure Monitor Call) API** for secure device authentication, leveraging **EFUSE key handlers** within the **ARM TrustZone** environment.

Engineer (Intern) - Airoha Jan 2024 – June 2024

- Developed and contributed **Linux drivers** for the **‘clk‘ (Clock)** and **‘watchdog‘ subsystems** that were successfully **upstreamed** to the mainline Linux Kernel repository.

PROJECTS

Upstream Linux Watchdog Driver [Link to Mainline Code](#)

Designed and implemented a robust **hardware-based Linux Watchdog driver** for a proprietary SoC, successfully merged into the **mainline Linux Kernel (v6.6+)**. The driver ensures platform resilience against system stalls and kernel panics by integrating with the **Common Clock Framework (CCF)** and **Reset Controller APIs**.

SoC Partition Scheme and TrustZone Verification Propriety Code

Designed and implemented a **GUID Partition Table (GPT)** based scheme to support dynamic partitioning for firmware updates. Integrated the verification logic within the **ARM Trusted Firmware (ATF)** at the **EL3/Secure Monitor** level to ensure integrity and secure the boot process.

OpenWRT Distribution Maintenance (21.02/23.05/24.10/25.12) Propriety Code

Led the maintenance and upgrade path from OpenWRT 21.02 to 23.05 and 24.10, resolving critical compilation and compatibility issues across the entire SoC software stack to maintain distribution stability.

EDUCATION

2019 - 2024 **Dual Degree (CSE) at NIT Hamirpur** (CGPA: 9.29/10.0)